

Homework 1

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1 *Nature* quantum survey

The AI classified me as a believer in many worlds, which (in a nutshell) states that the Schrödinger is behaved, and what we observe as a “collapse” of the wavefunction is us entangling ourselves with the system. The idea makes a lot of sense to me, and definitely makes more sense than the wavefunction just *collapsing*, but I’m not sure I fully buy it. It’s hard to really explain why, but I just don’t really believe it either.

2 The planck constant and other fundamental constants

- (i) Alfie is measuring the meter because c is fixed, and the meter is defined in terms of c .
 - (ii) Betty is measuring the kilogram. h is now defined, so by measuring $K = hf - \phi$, we are really measuring energy conservation, which is tied to the kilogram by relating energy to mass.
 - (iii) N_A is fixed, and the molar mass is simply mN_A for m the mass of one atom of that element. By counting the number of moles in a gram, since N_A is fixed, we are really measuring a gram in terms of the number of carbon atoms, thus measuring the *gram*.
 - (iv) Brownian motion experiments relate the mechanical energy of particles to the temperature via k_B . However, since k_B is fixed, we are really measuring temperature, hence the *kelvin*.
 - (v) The oil drop experiment relates macroscopic charge to the quantum of charge e , which is now fixed. Since the ampere is defined as $1/e$, we are really measuring the ampere.
- Basically all of these would be different if I had to answer pre-2018, since N_A, k_B, e, h were all fixed and the SI units recast in terms of them in 2018. Number (i) wouldn’t really change though.

3 Pauli spin matrices

(a)

$$\sigma_x \sigma_y = \begin{pmatrix} & 1 \\ 1 & \end{pmatrix} \begin{pmatrix} & -i \\ i & \end{pmatrix} = \begin{pmatrix} i & \\ & -i \end{pmatrix} = i\sigma_z.$$

$$[\sigma_y, \sigma_z] = \sigma_y \sigma_z - \sigma_z \sigma_y = \begin{pmatrix} & -i \\ i & \end{pmatrix} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} - \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \begin{pmatrix} & -i \\ i & \end{pmatrix} = \begin{pmatrix} & i \\ i & \end{pmatrix} - \begin{pmatrix} & -i \\ -i & \end{pmatrix} = 2i\sigma_x.$$

$$\sigma_x \sigma_y \sigma_z = i\sigma_z^2 = i \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} \begin{pmatrix} 1 & \\ & -1 \end{pmatrix} = i1.$$

(b) Obviously all Pauli matrices are Hermitian, and $\dagger$ distributes over addition, giving

$$\sigma_+^\dagger = (\sigma_x + i\sigma_y)^\dagger = \sigma_x^\dagger - i\sigma_y^\dagger = \sigma_x - i\sigma_y^\dagger = \sigma_-.$$

We also have

$$\sigma_-^\dagger = \sigma_+^{\dagger\dagger} = \sigma_+$$

since $(-)^{\dagger}$ is self-inverse. Therefore, these are clearly not Hermitian, since

$$\sigma_+ = \sigma_x + i\sigma_y = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \neq \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}^{\dagger} = \sigma_-.$$

We compute the eigenvalues for $\sigma_x, \sigma_y, \sigma_z$ in order:

$$\det \begin{pmatrix} -\lambda & 1 \\ 1 & -\lambda \end{pmatrix} = 0 \implies \lambda^2 - 1 = 0 \implies \lambda = \pm 1,$$

$$\det \begin{pmatrix} -\lambda & -i \\ i & -\lambda \end{pmatrix} = 0 \implies \lambda^2 - 1 = 0 \implies \lambda = \pm 1,$$

$$\det \begin{pmatrix} 1 - \lambda & 0 \\ 0 & -1 - \lambda \end{pmatrix} = 0 \implies (1 - \lambda)(-1 - \lambda) = 0 \implies -1 + \lambda^2 = 0 \implies \lambda = \pm 1.$$

We compute the eigenvectors for $\sigma_x, \sigma_y, \sigma_z$ in order:

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{pmatrix} 1 \\ 1 \end{pmatrix} \text{ e-vec for } \sigma_x,$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \implies \begin{pmatrix} y \\ x \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \implies x = -y \implies \begin{pmatrix} 1 \\ -1 \end{pmatrix} \text{ e-vec for } \sigma_x,$$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{pmatrix} -iy \\ ix \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies y = ix \implies \begin{pmatrix} 1 \\ i \end{pmatrix} \text{ e-vec for } \sigma_y,$$

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \implies \begin{pmatrix} -iy \\ ix \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \implies y = -ix \implies \begin{pmatrix} 1 \\ -i \end{pmatrix} \text{ e-vec for } \sigma_y,$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{pmatrix} x \\ -y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} \implies \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ e-vec for } \sigma_z,$$

$$\begin{pmatrix} x \\ -y \end{pmatrix} = \begin{pmatrix} -x \\ -y \end{pmatrix} \implies \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ e-vec for } \sigma_z.$$

(d) The following are well-known properties:

1. $\sigma_i^2 = 1$;
2. $\sigma_i \sigma_j = -\sigma_j \sigma_i$ for $i \neq j$;
3. $\sigma_i \sigma_j = \delta_{ij} 1 + \varepsilon_{ijk} i \sigma_k$ for ε_{ijk} the Levi-Civita symbol with the obvious ordering.

Proving these properties amounts to writing out a lot of different matrix equations, most of which contain only one step, and I don't think the grader would like to read that any more than I would like to write it.

In particular, we obtain

$$[\sigma_i, \sigma_j] = \sigma_i \sigma_j - \sigma_j \sigma_i = \varepsilon_{ijk} i \sigma_k + \varepsilon_{ijk} i \sigma_k = \varepsilon_{ijk} 2i \sigma_k.$$

We then have

$$\begin{aligned} [\sigma_x, [\sigma_y, \sigma_z]] + [\sigma_y, [\sigma_z, \sigma_x]] + [\sigma_z, [\sigma_x, \sigma_y]] &= [\sigma_x, 2i\sigma_x] + [\sigma_y, 2i\sigma_y] + [\sigma_z, 2i\sigma_z] \\ &= 2i([\sigma_x, \sigma_x] + [\sigma_y, \sigma_y] + [\sigma_z, \sigma_z]) \\ &= 0. \end{aligned}$$

4 Exponentials of operators

Recall matrix exponentials are defined by extending the Taylor series representation of the exponential

$$e^M := \sum_{n=0}^{\infty} \frac{M^n}{n!};$$

for M a matrix.

(a) (i) An easy induction argument shows that matrix exponentials of a diagonal matrix

$$\begin{pmatrix} m_1 & & \\ & \ddots & \\ & & m_n \end{pmatrix} = \text{diag}(m_1, \dots, m_n)$$

satisfy

$$\text{diag}(m_1, \dots, m_n)^k = \text{diag}(m_1^k, \dots, m_n^k).$$

Since $\sigma_z = \text{diag}(1, -1)$, we have

$$e^{i\theta\sigma_z} = \sum_{n=0}^{\infty} \frac{(i\theta\sigma_z)^n}{n!} = \sum_{n=0}^{\infty} \frac{\text{diag}(i\theta, -i\theta)^n}{n!} = \sum_{n=0}^{\infty} \frac{\text{diag}((i\theta)^n, (-i\theta)^n)}{n!}.$$

We factor the $n!$ into the matrix, and also note that by linearity of matrix addition we may also factor the sum into the matrix, giving

$$\sum_{n=0}^{\infty} \frac{\text{diag}((i\theta)^n, (-i\theta)^n)}{n!} = \sum_{n=0}^{\infty} \text{diag}\left(\frac{(i\theta)^n}{n!}, \frac{(-i\theta)^n}{n!}\right) = \text{diag}\left(\sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!}, \sum_{n=0}^{\infty} \frac{(-i\theta)^n}{n!}\right) = \text{diag}(e^{i\theta}, e^{-i\theta}),$$

where the last equality follows once again by noting the Taylor expansion of e^x .

(ii) For the rest of the problem, we will avoid explaining steps that were explained in (i) (for example, factoring the sum into diagonal matrices). Note the following:

$$\sigma_x^2 = 1 \implies \sigma_x^3 = \sigma_x.$$

In particular, $\sigma_x^n = \sigma_x^{n \pmod{2}}$. We will apply this fact to split the Taylor expansion around even and odd components:

$$\begin{aligned} e^{i\theta\sigma_x} &= \sum_{n=0}^{\infty} \frac{(i\theta\sigma_x)^n}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(i\theta\sigma_x)^{2n+1}}{n!} + \sum_{n=0}^{\infty} \frac{(i\theta 1)^{2n}}{(2n)!} = \sum_{n=0}^{\infty} (-1)^n \frac{i\theta^{2n+1}\sigma_x}{(2n+1)!} + \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n} 1}{(2n)!} \\ &= \left(\sum_{n=0}^{\infty} (-1)^n \frac{i\theta^{2n+1}}{(2n+1)!} \right) \sigma_x + \left(\sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n}}{(2n)!} \right) 1 = i \sin \theta \sigma_x + \cos \theta 1, \end{aligned}$$

where the final step follows by recognizing the Taylor expansions for $\sin$ and $\cos$. Factoring into the matrices gives the representation

$$\begin{pmatrix} & i \sin \theta \\ i \sin \theta & \end{pmatrix} + \begin{pmatrix} \cos \theta & \\ & \cos \theta \end{pmatrix} = \begin{pmatrix} \cos \theta & i \sin \theta \\ i \sin \theta & \cos \theta \end{pmatrix}.$$

(iii) We apply the same proof strategy employed in (ii) by recognizing that

$$\sigma_y^2 = 1 \implies \sigma_y^n = \sigma_y^{n \pmod{2}}.$$

It follows that

$$\begin{aligned}
e^{i\theta\sigma_y} &= \sum_{n=0}^{\infty} \frac{(i\theta\sigma_y)^n}{n!} = \sum_{n=0}^{\infty} \frac{(i\theta\sigma_y)^{2n+1}}{(2n+1)!} + \sum_{n=0}^{\infty} \frac{(i\theta 1)^{2n}}{(2n)!} = \sum_{n=0}^{\infty} (-1)^n \frac{i\theta^{2n+1}\sigma_y}{(2n+1)!} + \sum_{n=0}^{\infty} (-1)^n \frac{\theta^{2n} 1}{(2n)!} \\
&= i \sin \theta \sigma_y + \cos \theta 1 = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}.
\end{aligned}$$

(b) (i)

$$e^{i\theta\sigma_+} = \sum_{n=0}^{\infty} \frac{(i\theta)^n (\sigma_x + i\sigma_y)^n}{n!} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}^n = 1 + i\theta \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} + \sum_{n=2}^{\infty} \frac{(i\theta)^n}{n!} \mathbf{0} = \begin{pmatrix} 1 & 2i\theta \\ 0 & 1 \end{pmatrix}.$$

(ii)

$$e^{i\theta\sigma_i} = \sum_{n=0}^{\infty} \frac{(i\theta)^n (\sigma_x - i\sigma_y)^n}{n!} = \sum_{n=0}^{\infty} \frac{(i\theta)^n}{n!} \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix}^n = 1 + i\theta \begin{pmatrix} 0 & 0 \\ 2 & 0 \end{pmatrix} + \sum_{n=2}^{\infty} \frac{(i\theta)^n}{n!} \mathbf{0} = \begin{pmatrix} 1 & 0 \\ 2i\theta & 1 \end{pmatrix}.$$

(c) (i)

$$e^{-i\theta\sigma_+} = e^{i(-\theta)\sigma_+} = \begin{pmatrix} 1 & -2i\theta \\ 0 & 1 \end{pmatrix}.$$

$$\begin{pmatrix} 1 & 2i\theta \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & -2i\theta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -2i\theta \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & -2i\theta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & -4i\theta \\ 0 & -1 \end{pmatrix}.$$

(ii)

$$e^{-i\theta\sigma_-} \sigma_z e^{i\theta\sigma_+} = \begin{pmatrix} 1 & 0 \\ -2i\theta & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & 2i\theta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -2i\theta & -1 \end{pmatrix} \begin{pmatrix} 1 & 2i\theta \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 2i\theta \\ -2i\theta & 4\theta^2 - 1 \end{pmatrix}.$$

(d)

$$(\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) = a^i \sigma_i b^j \sigma_j = a^i b^j (\delta_{ij} 1 + \varepsilon_{ijk} i \sigma_k) = a^i b^i 1 + i a^i b^j \varepsilon_{ijk} \sigma_k = (\mathbf{a} \cdot \mathbf{b}) 1 + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}$$

by our results from section 2, and the fact that the k th component of $\mathbf{a} \times \mathbf{b} = a^i b^j \varepsilon_{ijk}$.

$$\begin{aligned}
[\mathbf{a} \cdot \boldsymbol{\sigma}, \mathbf{b} \cdot \boldsymbol{\sigma}] &= (\mathbf{a} \cdot \boldsymbol{\sigma})(\mathbf{b} \cdot \boldsymbol{\sigma}) - (\mathbf{b} \cdot \boldsymbol{\sigma})(\mathbf{a} \cdot \boldsymbol{\sigma}) = (\mathbf{a} \cdot \mathbf{b}) 1 + i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma} - ((\mathbf{b} \cdot \mathbf{a}) 1 + i(\mathbf{b} \times \mathbf{a}) \cdot \boldsymbol{\sigma}) \\
&= i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma} - i(\mathbf{b} \times \mathbf{a}) \cdot \boldsymbol{\sigma} = i((\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma} + (\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}) = 2i(\mathbf{a} \times \mathbf{b}) \cdot \boldsymbol{\sigma}.
\end{aligned}$$

5 Stern Gerlach experiment

(a) The control photograph shows the incidence locations of the silver atoms given no field. They show that in the presence of no field, the atoms do not deflect away from the center. This shows that any deflection must be due to the presence of a field.

(b) Not counting the spike at the top, it looks like the top and bottom are about 6 rulings apart, so ≈ 300 microns. This means the deflection from the center is 150 microns. Counting the spike, the

top deflects 300 microns away from the center.. To estimate the magnetic moment of the electron, we recall that

$$\mathbf{F} = \nabla(\boldsymbol{\mu} \cdot \mathbf{B}) \implies F_z = \mu_e \frac{\partial B_z}{\partial z}.$$

The atom is in the magnet of length L for a time $t = L/v_x$, for v_x x -velocity. Basic kinematics tells us the vertical displacement is

$$z = \frac{1}{2}a_z t^2.$$

Noting that $a_z = F_z/m$, substituting in gives

$$z = \frac{1}{2} \frac{\mu_e}{m} \frac{\partial B_z}{\partial z} \left(\frac{L}{v_x} \right)^2.$$

It follows that

$$\mu_e = \frac{2mzv_x^2}{L^2 \frac{\partial B_z}{\partial z}}.$$

To find v_x , note that the particle's speed is mostly in v_x , so we approximate the kinetic energy with the Maxwell speed distribution

$$\frac{1}{2}mv_x^2 \approx \frac{1}{2}k_B T \implies v_x \approx \sqrt{\frac{k_B T}{m}}.$$

Plugging in gives

$$\mu_e = \frac{2mzk_B T}{L^2 m \frac{\partial B_z}{\partial z}} = \frac{2zk_B T}{L^2 \frac{\partial B_z}{\partial z}}.$$

We have the following approximate values:

$$L = 0.035\text{m}, \quad \frac{\partial B_z}{\partial z} = 1000\text{T/m}, \quad z = 1.5 \cdot 10^{-4}\text{m},$$

$$T = 1273.15\text{K}, \quad k_B = 1.38 \cdot 10^{-23} \frac{\text{m}^2\text{kg}}{\text{s}^2\text{K}}.$$

Plugging in, we have

$$\mu_e = \frac{(2)(1.5 \cdot 10^{-4})(1.38 \cdot 10^{-23})(1.27315 \cdot 10^3)}{(0.035^2)(10^3)} \approx 4302.727 \cdot 10^{-27} = 4.302727 \cdot 10^{-24}.$$

This is in units of J/T. Also the magnetic moment of silver is the same as that of an electron because it has one valence electron, and other contributions cancel out.

(c) No, mass cancels out of the final equation.

(d) An electron is charged, and charged particles deflect when moving through $\mathbf{B}$ -fields. The electron charge would have messed up the numbers. Silver atoms cancel out the charge because there is a proton to each electron.

(e) Hopefully I found the right formula:

$$g_e = \frac{4m_e\mu_e}{e\hbar}.$$

Plugging in our findings, as well as accepted values for $m_e, e, \hbar$ give an approximate value of 0.928. That's not very close to the modern value.

(f) I don't know much about particle physics, but I assume it's because the muon is significantly more massive than the electron. This means that anomalies that might be related to the mass are easier to detect for the muon than the electron, since the mass scales much higher for the muon.

6 Pauli matrices as qubit operators

(a)

$$M_H = \frac{1}{\sqrt{2}} (\sigma_x + \sigma_z).$$

$$M_H|0\rangle = \frac{1}{\sqrt{2}} (\sigma_x + \sigma_z) |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{1}{\sqrt{2}} |0\rangle = \frac{1}{\sqrt{2}} (|1\rangle + |0\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = |+\rangle.$$

$$M_H|1\rangle = \frac{1}{\sqrt{2}} (\sigma_x + \sigma_z) |1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} - \frac{1}{\sqrt{2}} |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle.$$

The Hadamard gate puts a z -basis qubit into a superposition in the z -basis.

(b)

$$\det M_H = \frac{1}{2} (-1 - 1) = -1.$$

An inverse thus exists, using the regular formula for inverses for 2×2 s:

$$-\frac{1}{\sqrt{2}} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = M_H.$$

We verify the matrix is self-inverse:

$$\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = 1.$$

(c) (i) The probability is

$$|\langle \uparrow | \psi \rangle|^2 = \left| \frac{3}{\sqrt{34}} \langle \uparrow | \uparrow \rangle + i \frac{5}{\sqrt{34}} \langle \uparrow | \downarrow \rangle \right|^2 = \left| \frac{3}{\sqrt{34}} \right|^2 = \frac{9}{34}.$$

(ii) The expectation value is $\langle \psi | S_x | \psi \rangle$. Note that

$$S_x |\psi\rangle = \frac{\hbar}{2} \frac{1}{\sqrt{34}} \begin{pmatrix} 3 \\ 5i \end{pmatrix} = \frac{\hbar}{2\sqrt{34}} \begin{pmatrix} 5i \\ 3 \end{pmatrix}, \quad |\psi\rangle = \frac{1}{\sqrt{34}} \begin{pmatrix} 3 \\ 5i \end{pmatrix}.$$

Their inner product is thus

$$\frac{\hbar}{2 \cdot 34} (15i - 15i) = 0,$$

since

$$\langle \psi | = |\psi\rangle^\dagger = (3 \quad -5i).$$

(iii) The state is now collapsed to the eigenvector of S_x corresponding to -1 , which is simply that same eigenvector of σ_x since $S_x = \frac{1}{2}\sigma_x$. Thus, $|\psi\rangle$ is now

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} (|\uparrow\rangle - |\downarrow\rangle).$$

Since $|\psi\rangle$ is nonzero in both spin up and spin down, both possibilities of $\hbar/2$ and $-\hbar/2$ are measurable. The probabilities are obviously both $1/2$, but I compute them anyways:

$$|\langle \uparrow | \psi \rangle|^2 = \left| \frac{1}{\sqrt{2}} (\langle \uparrow | \uparrow \rangle - \langle \uparrow | \downarrow \rangle) \right|^2 = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}, \quad |\langle \downarrow | \psi \rangle|^2 = \left| \frac{1}{\sqrt{2}} (\langle \downarrow | \uparrow \rangle - \langle \downarrow | \downarrow \rangle) \right|^2 = \left| \frac{-1}{\sqrt{2}} \right|^2 = \frac{1}{2}.$$